

All You Need to Know About Snowball Products: Pricing, Risk Decomposition, and the Effect of Design Parameters*

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Abstract

Snowball structured products—autocallable notes with a down-and-in knock-in barrier—have become one of the most widely distributed retail derivatives in Asian markets, yet their risk profile is poorly understood by the investors who hold them. This paper provides a comprehensive pricing and risk analysis of the snowball under the [Bates \(1996\)](#) stochastic volatility with jumps (SVJ) model calibrated to S&P 500 data. We decompose the snowball payoff into its constituent implicit option positions—a short down-and-in put, a short call, and a long bond—and characterize the investor's implicit beliefs. We then study two design dimensions that issuers routinely vary: the knock-out (KO) barrier level and the KO observation frequency. We further benchmark all results against the market-standard Black–Scholes–Merton (BSM) model parameterized by the ATM implied volatility. Our key findings are fourfold. First, raising the

*The author thanks seminar participants for helpful comments. All errors are my own.

KO barrier from 103% to 105% of the initial spot materially increases the knock-in (KI) loss probability (from 14.9% to 18.1%) while offering only marginal gains in full-coupon probability. Second, more frequent KO observations lower the product's fair value despite increasing the probability of early exit, because coupon accrual is curtailed; they simultaneously reduce knock-in risk by rescuing upward-trending paths before they can crash through the barrier. Third, the product's delta profile peaks sharply near the knock-in barrier and collapses above the knock-out level, generating the negative convexity that drives the well-documented "snowball effect" in dealer hedging books. Fourth, despite matching the ATM vanilla option price by construction, the BSM-IV benchmark systematically underestimates the autocall probability (82.0% vs. 83.8%) and overestimates the knock-in loss probability (16.6% vs. 14.9%) relative to the Bates SVJ model. Stochastic volatility clustering generates high-vol episodes that push paths through the low 103% KO barrier more frequently, creating a rescue effect that reduces KI exposure; flat-vol BSM, unable to replicate these surges, overestimates residual tail risk with direct implications for hedge accuracy.

Keywords: snowball, autocallable, structured products, Bates model, stochastic volatility, Monte Carlo, barrier options, delta hedging.

JEL: G12, G13, C15.

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1 Introduction

Structured products that combine a high coupon with barrier-contingent capital protection have attracted enormous retail interest, particularly in China and South Korea, where snowball-type notes reportedly accounted for hundreds of billions of dollars in outstanding notional by the early 2020s. The appeal is intuitive: in a low-interest-rate environment, a 15–25% annual coupon on an equity-linked note appears attractive, especially when accompanied by the assurance that capital is at risk only if the underlying index falls by more than 20–30%.

What many investors fail to appreciate is that this coupon is not a free lunch. It is the premium they receive for writing two implicit options simultaneously: a *down-and-in put* that exposes them to full equity downside if the underlying breaches the knock-in barrier at any point during the product’s life, and a *call* that is effectively sold to the issuer in the form of the knock-out autocall feature, which terminates the product and caps the investor’s upside whenever the underlying trades above the KO barrier on a scheduled observation date. The snowball investor is therefore a leveraged short-volatility trader dressed in the clothing of a high-yield bond investor.

This misalignment between perception and economic reality motivates the present study. We pursue four contributions. First, we derive a transparent option-based decomposition of the snowball payoff and characterize precisely what risk premia the investor collects. Second, we use Monte Carlo simulation under the [Bates \(1996\)](#) SVJ model—which captures the fat tails, negative skewness, and stochastic volatility of equity index returns—to quantify how design choices (KO barrier level and observation frequency) affect the product’s fair value, scenario probabilities, and sensitivity profile. Third, we document the delta structure of the snowball and discuss its implications for dealer risk management. Fourth, we benchmark the Bates SVJ results against the Black–Scholes–Merton model parameterized by the market ATM implied volatility, isolating the contribution of stochastic volatility and jump dynamics to snowball pricing and hedge errors.

The remainder of the paper is organized as follows. Section 2 describes the product structure. Section 3 presents the modeling framework. Section 4 details the Monte Carlo pricing methodology. Section 5 derives the option-bundle decomposition and the investor's implicit beliefs. Sections 6 and 7 analyze the effects of the KO barrier level and observation frequency, respectively. Section 8 benchmarks the Bates SVJ results against BSM with ATM implied volatility. Section ?? discusses risk management implications. Section 11 concludes.

2 The Snowball Product

2.1 Payoff Structure

A snowball structured product is a note issued at par (face value normalized to 1) on an underlying asset with initial price S_0 . It has a fixed maturity T , a coupon rate c per annum, a knock-out barrier at level $B_{KO} = \kappa_{KO}S_0$, and a knock-in barrier at level $B_{KI} = \kappa_{KI}S_0$, where $\kappa_{KO} > 1 > \kappa_{KI}$.

Stopping times. Let $\mathcal{T} = \{t_1, t_2, \dots, t_n\}$ with $t_1 < t_2 < \dots < t_n = T$ denote the scheduled KO observation dates. Define the *knock-out time*

$$\tau_{KO} = \inf\{t_i \in \mathcal{T} : S_{t_i} \geq B_{KO}\} \quad (1)$$

(with the convention $\inf \emptyset = +\infty$) and the *knock-in first hitting time*

$$\tau_{KI} = \inf\{t \in [0, T] : S_t \leq B_{KI}\} \quad (2)$$

(with the convention $\inf \emptyset = +\infty$). The knock-out is monitored only on the discrete grid $\mathcal{T}$, while the knock-in barrier is monitored *continuously*, so τ_{KI} is the classical first hitting time of the level B_{KI} from above.

Complete payoff. The three events $\{\tau_{\text{KO}} \leq T\}$, $\{\tau_{\text{KO}} > T, \tau_{\text{KI}} \leq T\}$, and $\{\tau_{\text{KO}} > T, \tau_{\text{KI}} > T\}$ partition the sample space almost surely. The investor's discounted payoff is

$$V = \begin{cases} e^{-r\tau_{\text{KO}}}(1 + c\tau_{\text{KO}}) & \text{if } \tau_{\text{KO}} \leq T, \\ e^{-rT} \min\left(\frac{S_T}{S_0}, 1\right) & \text{if } \tau_{\text{KO}} > T \text{ and } \tau_{\text{KI}} \leq T, \\ e^{-rT}(1 + cT) & \text{if } \tau_{\text{KO}} > T \text{ and } \tau_{\text{KI}} > T. \end{cases} \quad (3)$$

In the first case the note autocalls at τ_{KO} ; the investor receives principal plus accrued coupon. In the second case the product survives to maturity but the KI barrier has been breached at some point: if $S_T < S_0$ the investor bears the full proportional loss (S_T/S_0), while if $S_T \geq S_0$ the asset has recovered and the investor receives par but forfeits the coupon. In the third case neither barrier event has occurred and the investor earns the full coupon over the entire tenor.

2.2 Economic Intuition

The product offers three possible outcomes:

1. **Early autocall (KO triggered).** The most common outcome in rising or stable markets. The investor receives principal plus accrued coupon, generating a high annualized internal rate of return.
2. **Capital loss (KI triggered, no KO).** The downside scenario. The investor effectively holds the underlying from S_0 , absorbing any loss incurred below the KI barrier.
3. **Full coupon at maturity (neither KO nor KI).** The least frequent outcome. The market neither rallied enough to autocall nor fell far enough to knock in; the investor earns the full coupon over the entire tenor.

In our baseline calibration ($\kappa_{\text{KO}} = 1.03$, $\kappa_{\text{KI}} = 0.75$, $c = 20\%$, $T = 2$ years, monthly

observations), these three scenarios occur with probabilities approximately 84%, 15%, and 1%, respectively, under the Bates SVJ model.

3 Modeling Framework

3.1 The Bates Stochastic Volatility with Jumps Model

We adopt the [Bates \(1996\)](#) model as the data-generating process. Under the risk-neutral measure, the joint dynamics of the asset price S_t and instantaneous variance v_t are

$$\frac{dS_t}{S_t} = (r - q - \lambda_J \bar{\mu}_J) dt + \sqrt{v_t} dW_t^S + (e^{Y_t} - 1) dN_t, \quad (4)$$

$$dv_t = \kappa(\theta - v_t) dt + \sigma_v \sqrt{v_t} dW_t^v, \quad (5)$$

where W^S and W^v are standard Brownian motions with $d\langle W^S, W^v \rangle_t = \rho dt$; N_t is a Poisson process with constant intensity λ_J ; $Y_t \sim \mathcal{N}(\mu_J, \delta_J^2)$ represents the log-jump size; and $\bar{\mu}_J = e^{\mu_J + \delta_J^2/2} - 1$ is the compensator that ensures S_t is a martingale after discounting.

The parameters have natural interpretations: κ is the speed of mean reversion of variance to its long-run level θ ; σ_v is the volatility of volatility; $\rho < 0$ captures the leverage effect; λ_J is the expected number of jumps per year; and $\mu_J < 0$ and $\delta_J > 0$ describe the typically negative and significant jump-size distribution for equity indices.

The Bates model nests the [Heston \(1993\)](#) stochastic-volatility model (setting $\lambda_J = 0$) and the [Merton \(1976\)](#) jump-diffusion model (setting $\sigma_v = 0$), and is therefore well-suited for reproducing the steep short-maturity implied-volatility skew and fat-tailed return distribution observed in S&P 500 options.

3.2 Parameter Calibration

We calibrate the model to S&P 500 implied-volatility data using parameter values representative of a normal market regime. The calibrated parameters are reported in [Table 1](#). The

initial asset level is set to $S_0 = 4,500$, the risk-free rate to $r = 2\%$, and the dividend yield to $q = 0\%$.

Table 1: Bates SVJ model parameters (S&P 500 calibration)

Parameter	Symbol	Value	Interpretation
Mean reversion speed	κ	2.00	Variance half-life ≈ 4 months
Long-run variance	θ	0.05	Long-run annualized vol $\approx 22\%$
Volatility of variance	σ_v	0.30	Moderate vol-of-vol
Return–variance corr.	ρ	−0.70	Strong leverage effect
Initial variance	v_0	0.05	Current annualized vol $\approx 22\%$
Jump intensity	λ_J	0.12	Expected 1 jump per ≈ 8 years
Mean log-jump size	μ_J	−0.05	Average jump $\approx -5\%$
Jump size volatility	δ_J	0.25	Significant jump uncertainty

4 Monte Carlo Pricing

4.1 Discrete-Time Simulation

We simulate the Bates SVJ dynamics using an Euler–Maruyama discretization with a daily time step $\Delta t = T/N$ where $N = 504$ (252 trading days per year over two years). For path i and time step k :

$$v_{k+1}^{(i)} = \max\left(v_k^{(i)} + \kappa\left(\theta - v_k^{(i)}\right) \Delta t + \sigma_v \sqrt{v_k^{(i)}} \Delta t Z_{v,k}^{(i)}, 0\right), \quad (6)$$

$$S_{k+1}^{(i)} = S_k^{(i)} \exp\left[\left(r - q - \lambda_J \bar{\mu}_J - \frac{1}{2} v_k^{(i)}\right) \Delta t + \sqrt{v_k^{(i)}} \Delta t Z_{S,k}^{(i)}\right] J_k^{(i)}, \quad (7)$$

where $Z_{v,k}^{(i)}$ and $Z_{S,k}^{(i)}$ are correlated standard normal draws with $\text{Corr}(Z_S, Z_v) = \rho$, and $J_k^{(i)} = \exp(\sum_{l=1}^{M_k} Y_l)$ is the compound jump factor with $M_k \sim \text{Poisson}(\lambda_J \Delta t)$. The variance process is reflected at zero to enforce non-negativity.

All results are based on 50,000 simulated paths. Sensitivity curves (PV and delta as functions of the current spot) are computed using *common random numbers*: a single set of 50,000 standard-normal arrays is drawn once, and paths are scaled by the current spot

level, so that differences across spot values and design parameters are attributable entirely to structural variation rather than simulation noise.

4.2 Delta Estimation

Delta is estimated numerically by computing the snowball's fair value at 50 evenly spaced spot levels in $[0.55 S_0, 1.45 S_0]$ and applying a centered finite difference:

$$\hat{\Delta}(S) \approx \frac{V(S + \delta S) - V(S - \delta S)}{2 \delta S}. \quad (8)$$

This approach produces smooth, low-noise delta curves without additional path count.

5 The Snowball as an Implicit Option Bundle

5.1 Payoff Decomposition

The snowball's simplified at-maturity payoff (abstracting from path-dependent early termination for expositional clarity) can be written as

$$V_T = (1 + cT) - \underbrace{\left[(1 + cT) - \frac{S_T}{S_0} \right] \mathbf{1}_{\{S_T < B_{KI}\}}}_{\text{knock-in loss}}. \quad (9)$$

The first term is a risk-free bond paying $1 + cT$. The second term is the loss the investor sustains when the KI is triggered, which equals $(1 + cT) - S_T/S_0$ below B_{KI} and zero otherwise. This loss decomposes further as

$$\left[(1 + cT) - \frac{S_T}{S_0} \right] \mathbf{1}_{\{S_T < B_{KI}\}} = \underbrace{(1 + cT - \kappa_{KI}) \mathbf{1}_{\{S_T < B_{KI}\}}}_{\text{digital put at } B_{KI}} + \underbrace{\left(\kappa_{KI} - \frac{S_T}{S_0} \right)^+}_{\text{vanilla put at } B_{KI}}. \quad (10)$$

The snowball is therefore equivalent to a long bond minus a short down-and-in digital put

minus a short down-and-in vanilla put, all evaluated at the KI barrier. The autocall feature additionally implies that the investor has sold a call on the upside beyond B_{KO} : whenever the underlying reaches the KO level, the investor's payoff is capped at $1 + ct_i$ and the product terminates.

Proposition 1. *The fair value of a snowball with parameters $(S_0, B_{KO}, B_{KI}, c, T, r)$ equals the value of a portfolio consisting of:*

1. *a long position in a risk-free bond maturing at T with face value $1 + cT$;*
2. *a short position in a down-and-in digital put with barrier B_{KI} and rebate $1 + cT - \kappa_{KI}$;*
3. *a short position in a down-and-in vanilla put struck at B_{KI} with knock-in barrier B_{KI} ;*
4. *a short Bermudan call with observation dates $\{t_i\}$ and strike B_{KO} , reflecting the forgone upside above the KO level.*

The coupon c is the aggregate premium that prices this portfolio at par. **The snowball investor holds a synthetic high-yield bond whose coupon is the premium received for writing a barrier put and a capped call.**

5.2 The Three Payoff Zones

Figure 2 illustrates this decomposition across three distinct zones.

Below the KI barrier ($S_T < B_{KI}$): the short-put zone. The investor receives S_T/S_0 , the proportional value of the underlying. The loss relative to par exceeds that of a simple vanilla put struck at B_{KI} because the investor also forfeits the entire coupon. Panel (b) of Figure 2 overlays the snowball payoff against a short vanilla put plus bond, showing that the snowball suffers a sharper loss below the KI barrier due to this additional digital component.

Between the barriers ($B_{KI} \leq S_T \leq B_{KO}$): the bond zone. The investor collects the full coupon $1 + cT$, regardless of where within the range the underlying trades at maturity. This is the investor’s desired scenario.

Above the KO barrier ($S_T > B_{KO}$): the short-call zone. The product would have autocalled before maturity. The investor’s upside is capped at the accrued coupon at the time of autocall; any rally beyond B_{KO} is foregone. Panel (c) of Figure 2 overlays the snowball payoff against a long-stock benchmark, shading the forgone upside above the KO level.

5.3 Investor’s Implicit Beliefs

An investor who purchases a snowball at par must implicitly hold the following views to expect a positive expected profit.

1. **Low probability of a large drawdown.** The investor expects the probability of the underlying falling more than $1 - \kappa_{KI} = 25\%$ from its initial level at any point during the product’s life to be small. Under our Bates SVJ calibration with monthly KO monitoring, this probability is approximately 14.9% for $\kappa_{KO} = 1.03$ and 18.1% for $\kappa_{KO} = 1.05$ —not negligible, and materially larger than what investors typically assume under a diffusion-only model.
2. **Moderate bullishness or range-bound market view.** The investor expects the underlying to remain in the $[\kappa_{KI}, \kappa_{KO}]$ corridor for a meaningful fraction of the product’s life, or to drift above B_{KO} and trigger an autocall early, generating a high annualized return.
3. **Comfort with selling volatility.** The high coupon compensates the investor for being short a strangle (a barrier put plus a capped call). Investors who do not recognize this implicit short-volatility position may be surprised when large market moves, in either direction, work against them.

6 Effect of the KO Barrier Level

6.1 Scenario Probabilities

Table 2 compares the snowball with $\kappa_{\text{KO}} = 1.03$ and $\kappa_{\text{KO}} = 1.05$, holding all other parameters fixed.

Table 2: Scenario probabilities by KO barrier level

	KO = 103%	KO = 105%	Change
P(KO early exit)	83.8%	79.7%	−4.1 pp
P(KI capital loss)	14.9%	18.1%	+3.2 pp
P(full coupon)	1.3%	2.2%	+0.9 pp

Monthly observation, KI = 75%, coupon = 20%/yr, T = 2yr, Bates SVJ, 50,000 paths.

Raising the KO barrier from 103% to 105% reduces the autocall probability by about 4 percentage points. Because these paths no longer exit early through the KO, they remain exposed throughout the product’s life, and a meaningful fraction eventually triggers the KI barrier. As a result, the KI loss probability rises from 14.9% to 18.1%—an increase of 3.2 percentage points, or roughly 21% in relative terms.

From the investor’s perspective, a higher KO level appears more attractive because the product is “harder to call away.” However, this perception is misleading: the additional time in the product corresponds to additional exposure to the KI tail risk. The narrow gain in full-coupon probability (0.9 percentage points) does not compensate for the commensurate increase in loss probability.

6.2 Cumulative KO Probability and Timing

Figure 4 shows that the cumulative KO probability follows a concave increasing curve over the 24 monthly observation dates. At the 6-month mark, $\kappa_{\text{KO}} = 1.03$ has already triggered approximately 69% of its final KO paths, compared to 61% for $\kappa_{\text{KO}} = 1.05$. The slower pace of autocalls under the higher KO means the investor holds the product longer on average,

increasing both coupon accrual (beneficial) and KI exposure (harmful).

6.3 PV and Delta Profiles

Both KO configurations price close to par at the initial spot, confirming that the 20% annual coupon is approximately fair compensation for the risk borne at monthly observation frequency. The PV curves differ in the region just below the KO barrier (Figure 5, upper panel): the $\kappa_{\text{KO}} = 1.05$ product has a slightly higher PV in the $[B_{\text{KO},103\%}, B_{\text{KO},105\%}]$ corridor because paths in that range are not yet autocalled, and coupon continues to accrue.

The delta profiles (Figure 5, lower panel) share the same qualitative shape—peak sensitivity near the KI barrier, declining to near zero above the KO level—but the $\kappa_{\text{KO}} = 1.05$ delta peak is shifted approximately 20 basis points to the right, tracking the higher autocall threshold.

7 Effect of KO Observation Frequency

7.1 Motivation

In practice, issuers offer snowballs with KO monitoring ranging from annual to weekly. The observation schedule is a first-order design choice: more frequent checks give the investor more opportunities to exit early but also accelerate the coupon clock. This section quantifies the tradeoff.

7.2 Pricing Results

Table 3 reports the snowball’s fair value at $S = S_0$ and scenario probabilities under four monitoring schedules.

Two findings stand out.

Table 3: Effect of KO observation frequency on pricing and scenario probabilities

Frequency	Obs. dates	P(KO)	P(KI)	P(full coupon)	PV at S_0
Annual	2	61.3%	30.6%	8.1%	1.0528
Quarterly	8	77.0%	20.5%	2.6%	1.0091
Monthly	24	83.8%	14.9%	1.3%	0.9997
Weekly	104	88.2%	11.0%	0.7%	0.9961

KO = 103%, KI = 75% continuously monitored, coupon = 20%/yr, T = 2yr, Bates SVJ, 50,000 paths.

Finding 1: More frequent KO observation reduces the product’s fair value. The PV falls monotonically from 1.053 (annual) to 0.996 (weekly). This result is initially counterintuitive: more opportunities to autocall seems like a better deal for the investor. The resolution is that $e^{-rt_i}(1 + ct_i) \approx 1 + (c - r)t_i$ is increasing in t_i when $c > r$ (as here, since $c = 20\% \gg r = 2\%$). By triggering autocalls earlier, more frequent monitoring curtails the time over which the investor earns the premium $(c - r) = 18\%$ per year. The loss of coupon accrual dominates the benefit of early cash receipt.

Finding 2: More frequent KO observation materially reduces the knock-in loss probability. P(KI) falls from 30.6% under annual monitoring to 11.0% under weekly monitoring—a reduction of nearly 20 percentage points. The mechanism is path-dependent: upward-trending paths that would have survived to eventually crash through the KI barrier under infrequent monitoring are “rescued” by an earlier KO trigger when monitoring is frequent. In other words, frequent observation acts as a partial hedge against the KI tail risk by converting potentially dangerous surviving paths into early autocalls before the downside scenario materializes.

These two findings imply a **frequency-dependent risk-return tradeoff** for the investor: lower monitoring frequency implies higher PV (the product is worth more because it accrues more coupon) but also higher KI loss probability. Investors and issuers must weigh these opposing forces when structuring the observation schedule.

Remark 1. *The monthly schedule ($PV = 0.9997 \approx 1$) prices essentially at par under our*

calibration, suggesting that 20%/yr is approximately the fair coupon for monthly observation. Annual monitoring produces an above-par PV of 1.053, implying the investor receives excessive compensation for the risk borne; weekly monitoring produces a below-par PV of 0.996, implying the coupon is slightly insufficient at that frequency. Fair pricing therefore requires explicit agreement between issuer and investor on the observation schedule.

7.3 Delta Sensitivity by Observation Frequency

Figure 7 overlays the delta curves for the four monitoring schedules. The key features are:

1. **Peak delta near the KI barrier.** All four schedules share a delta peak in the vicinity of $S = 0.75 S_0$, reflecting that this is the region where the KI barrier is most likely to be triggered by a small additional downward move.
2. **Higher, narrower peak under more frequent monitoring.** The weekly delta curve exhibits a taller, more concentrated peak near the KO level ($S \approx B_{\text{KO}}$) compared to the annual curve. With frequent observation dates, the probability of triggering a KO jumps sharply when the spot crosses B_{KO} on any given observation date, creating a steep gradient in PV that manifests as a large delta.
3. **Faster delta decay above the KO barrier.** Under frequent monitoring, paths above B_{KO} are almost certain to autocall in the near future; PV becomes nearly flat there, implying near-zero delta. Under annual monitoring, the spot can drift above B_{KO} between observation dates without triggering autocall, so PV remains more sensitive to spot movements above the KO level.

8 BSM with Implied Volatility as Market Benchmark

8.1 Methodology

Market practitioners who price and hedge autocallable products routinely employ the Black–Scholes–Merton (BSM) model parameterized by the option-market implied volatility. Rather than a historical or long-run realized volatility, the input is the single flat volatility—anchored at the ATM implied vol at the relevant tenor—that is consistent with the current option market. This approach constitutes the industry benchmark: it is computationally simple, market-calibrated in the sense that it matches the ATM vanilla option price, and widely used for marking books and generating hedge ratios.

To construct this benchmark we extract the 2-year ATM implied volatility from the Bates SVJ model by pricing a 2-year ATM European call via Monte Carlo (100,000 paths) and inverting the Black–Scholes formula:

$$\sigma_{IV} = \text{BSM}^{-1}(C^{\text{Bates}}(S_0, K = S_0, T = 2)) = 23.48\%. \quad (11)$$

Using $\sigma_{IV} = 23.48\%$ as the sole volatility input, we then simulate 50,000 paths under constant-volatility GBM dynamics:

$$S_{t+\Delta t} = S_t \exp \left[\left(r - q - \frac{1}{2} \sigma_{IV}^2 \right) \Delta t + \sigma_{IV} \sqrt{\Delta t} Z_t \right], \quad (12)$$

and apply the identical snowball payoff function to price the product and compute PV and delta curves. By construction, both the Bates SVJ model and the BSM-IV model agree on the fair value of the 2-year ATM vanilla call; all differences in the snowball price and Greeks are attributable entirely to the higher-order distributional features—smile, skew, stochastic volatility, and jump risk—that BSM cannot capture.

8.2 Scenario Probabilities and Pricing

Table 4 reports the scenario probabilities and fair value under both models.

Table 4: Bates SVJ vs. BSM-IV benchmark

	Bates SVJ	BSM-IV ($\sigma_{IV} = 23.48\%$)
P(KO early exit)	83.8%	82.0%
P(KI capital loss)	14.9%	16.6%
P(full coupon)	1.3%	1.3%
PV at S_0	1.0032	1.0024

KO = 103%, KI = 75%, coupon = 20%/yr, T = 2yr, monthly obs., 50,000 paths.

Despite sharing the same ATM vanilla price, the two models produce systematically different scenario probabilities. The Bates SVJ autocall probability is 1.8 percentage points higher (83.8% vs. 82.0%), while its knock-in loss probability is 1.7 percentage points lower (14.9% vs. 16.6%).

The mechanism is **stochastic volatility clustering**. In the Bates model, instantaneous variance v_t evolves stochastically around its long-run level θ , generating episodes in which annualized volatility temporarily reaches 30–40%. During these high-vol surges, the underlying moves more violently, making it substantially easier to breach the relatively modest 103% KO barrier on any observation date. The consequence is a **rescue effect**: volatile paths that would, under a calmer trajectory, drift downward toward the KI barrier are instead terminated early via autocall, removing them from the pool of paths with ongoing KI exposure. Because BSM applies a fixed $\sigma_{IV} = 23.48\%$ —effectively the average volatility—it cannot replicate these episodic surges. Fewer paths are rescued by early KO exits, and a larger fraction survives long enough to risk breaching the KI barrier.

The net effect on PV is small in absolute terms: Bates SVJ prices the product at 1.0032 versus BSM-IV’s 1.0024, a difference of 0.0008 per unit of principal. The opposing forces—higher KO frequency (beneficial to value) and lower KI frequency (also beneficial) under Bates—nearly offset each other at the product level. However, the divergence in scenario probabilities is substantial at the hedge level: a dealer relying on BSM-IV will overestimate

KI exposure by approximately 11% in relative terms and underestimate the pace of early termination accordingly.

8.3 PV and Delta Profiles

Figure 8 overlays the PV and delta curves for both models. The PV curves are nearly indistinguishable across most of the spot range, diverging modestly in the $[B_{KI}, B_{KO}]$ corridor where distributional differences matter most. The Bates SVJ PV curve lies slightly above the BSM-IV curve throughout this region, consistent with the higher frequency of early, coupon-bearing autocalls.

The delta profiles reveal a more consequential difference. The Bates SVJ delta peak is more concentrated—taller and narrower—than the BSM-IV peak. Under stochastic volatility, the gamma exposure is more heavily focused in the region just above the KI barrier: during high-vol regimes, a small downward move has an outsized effect on KI-crossing probability, producing a sharper PV gradient. The BSM-IV delta is smoother and its peak more diffuse, understating the true gamma concentration that dealers will face precisely in the market conditions—elevated realized volatility, sharp downside moves—when the KI scenario is most imminent. A delta hedge computed from the BSM-IV model will therefore be imprecise in the regimes where accurate hedging matters most.

9 Information Asymmetry and the Implied Volatility Surface

9.1 The Implied Volatility Surface as an Incomplete Statistic

Financial institutions that structure and sell snowball products typically possess the full Bates SVJ parameter vector $\Theta = (\kappa, \theta, \sigma_v, \rho, v_0, \lambda_J, \mu_J, \delta_J)$, estimated from a broad cross-section of vanilla option prices, variance swap rates, realised variance time series, and pro-

prietary historical data. Retail investors—and many institutional buyers—have access only to the publicly observable *implied volatility surface*. This information gap has direct pricing implications.

The mapping from Bates parameters to implied volatilities is many-to-one. Define the equivalence class of the true parameter vector Θ^* at calibration tolerance ε as

$$\mathcal{I}(\Theta^*) = \{\Theta : \|\mathcal{S}(\Theta) - \mathcal{S}(\Theta^*)\|_\infty \leq \varepsilon\}, \quad (13)$$

where $\mathcal{S}(\Theta) = \{\sigma_{\text{BS}}(K_j, T_j; \Theta)\}$ is the Black–Scholes implied volatility surface generated by Θ . Vanilla option prices constrain only the marginal distributions of S_T at each maturity; the snowball payoff, by contrast, depends on the joint distribution of the entire price path—in particular, on $\min_{t \in [0, T]} S_t$ and on $S_{t_1}, \dots, S_{t_n}$ —which is not uniquely determined by $\mathcal{S}(\Theta)$. This non-identifiability has been documented rigorously for accumulator contracts (Suo, 2026), where parameter sets with materially different spot-variance correlations ρ match the same surface to within calibration tolerance yet price the path-dependent product at systematically different levels.

9.2 Which Parameters Drive Snowball Non-Identifiability?

Two Bates parameters are especially consequential for snowball pricing and especially difficult to identify from the implied volatility surface alone.

Spot-variance correlation ρ . The parameter ρ is the primary driver of the implied volatility skew because it controls the marginal distribution of S_T through the leverage effect. But ρ also governs the *joint* path dynamics of (S_t, v_t) : specifically, how violently variance surges when the spot falls, and therefore how often a downward-trending path is rescued by a subsequent high-variance episode that pushes the spot back through the KO barrier. The rescue effect documented in Section 8 is driven by this mechanism. For a given skew target, multiple (ρ, κ, σ_v) combinations can match the same implied volatility surface—

a less negative ρ can be compensated by a higher σ_v or a different κ —with marginal impact on observed vanilla option prices but potentially large impact on the snowball’s KI and KO probabilities.

Volatility-of-volatility σ_v and mean-reversion speed κ . The implied volatility surface constrains certain combinations of σ_v and κ through the term structure of implied variance, but individual values can differ substantially across surface-equivalent parameter sets. A model with lower κ allows high-variance episodes to persist for longer; one with higher σ_v generates more extreme variance spikes. Both features affect the depth and duration of the volatility clustering responsible for the rescue mechanism, yet the implied volatility surface cannot separately identify them.

9.3 Investor Overpricing and the Seller’s Informational Rent

The seller sets the coupon c^* to make the snowball zero-NPV at inception under the true dynamics Θ^* , that is, $V_0(\Theta^*, c^*) = 0$. An investor who calibrates to a surface-equivalent but incorrect $\Theta^{\text{inv}} \in \mathcal{I}(\Theta^*)$ evaluates the product at

$$V_0^{\text{inv}}(c^*) = V_0(\Theta^{\text{inv}}, c^*), \quad (14)$$

which generally differs from zero. The *seller’s informational rent* is $-V_0^{\text{inv}}(c^*)$: when positive, the investor believes they are entering a positive-NPV transaction at c^* , and the seller can marginally reduce the coupon without the investor’s model flagging the product as unattractive.

The direction of the mispricing depends on whether Θ^{inv} over- or underestimates the rescue effect—but, importantly, this is *not* pinned down by the sign of $\rho^{\text{inv}} - \rho^*$ alone. Because ρ enters the equivalence class $\mathcal{I}(\Theta^*)$ only jointly with a compensating adjustment to another parameter— σ_v or κ —that restores the observed implied volatility surface, the rescue effect responds to the *combination* of parameters that is varied, not to ρ in isolation.

We illustrate this with a direct numerical experiment rather than relying on intuition alone.

Numerical illustration. Holding $\kappa, \theta, v_0, \lambda_J, \mu_J, \delta_J$ at the baseline values of Table 1, we solve by Monte Carlo for the true model’s zero-NPV coupon, $c^* \approx 19.8\%$ —close to, and consistent with, the baseline $c = 20\%$ used elsewhere in the paper, confirming that 20% prices essentially at par. We then perturb ρ to $\rho^{\text{inv}} \in \{-0.85, -0.55\}$ and re-solve for σ_v^{inv} , by bisection on the same Monte Carlo engine, so that the resulting 2-year ATM implied volatility matches the true model’s (23.4%) to within numerical tolerance—the same single-point surface-matching standard used for the BSM-IV benchmark of Section 8. A single consistent random seed is used for both the calibration target and the bisection, eliminating cross-seed Monte Carlo noise in the objective function. Each Θ^{inv} constructed this way is observationally indistinguishable from Θ^* at the ATM point, yet the two parameter sets imply different snowball valuations. Table 5 reports the results (500,000 paths, common random numbers across all parameter sets).

Table 5: Numerical magnitude of the seller’s informational rent under surface-equivalent re-parameterizations

	ρ^{inv}	σ_v^{inv}	P(KI loss)	$V_0^{\text{inv}}(c^*)$	Coupon-equiv. gap
True model (Θ^*)	−0.70	0.300	15.2%	0 (by constr.)	—
More negative ρ^{inv}	−0.85	0.277	15.2%	−0.0013	−44 bp
Less negative ρ^{inv}	−0.55	0.321	15.2%	+0.0017	+56 bp

$c^* = 19.7\%$, $KO = 103\%$, $KI = 75\%$, $T = 2\text{yr}$, *monthly observation, 2-year ATM implied vol matched to 23.4% in all rows. $V_0^{\text{inv}}(c^*)$ is per unit of principal; e.g. −0.0013 corresponds to −\$13,000 per \$10 million notional.*

Counter-intuitively, it is the *less negative* ρ^{inv} —paired with its compensating $\sigma_v^{\text{inv}} = 0.321$ —that produces the *positive* $V_0^{\text{inv}}(c^*)$: the investor overprices the product, and the seller can extract a rent of roughly 56 basis points of annual coupon (about \$17,000 per \$10 million of notional at this calibration) by quoting a coupon the investor’s model will still flag as fair. The more negative ρ^{inv} , paired with its own compensating $\sigma_v^{\text{inv}} = 0.277$, produces the *opposite* sign: the investor underprices the product and would, if anything,

demand a higher coupon than the seller needs to offer. This is the reverse of what a naive reading of $\text{sign}(\rho^{\text{inv}} - \rho^*)$ alone would predict. The lesson is that the *magnitude* of the rent is modest under a single-parameter compensation—on the order of 44–56 basis points of coupon, or 0.13–0.17% of notional, for a ± 0.15 shift in ρ —but its *sign* depends on which parameter in Θ absorbs the compensation required to preserve the observed surface, not on ρ^{inv} in isolation. This strengthens, rather than weakens, the case for disclosure made below: an investor cannot infer even the *direction* of their pricing error from the sign of their own assumed leverage effect.

A structurally larger and directionally unambiguous example is the BSM-IV benchmark of Section 8, which corresponds to the polar case $\sigma_v^{\text{inv}} \rightarrow 0$ —eliminating stochastic volatility, and hence the rescue effect, altogether rather than merely reweighting it. There, the reported KI probabilities (16.6% vs. 14.9%) and prices (1.0024 vs. 1.0032) imply, using the coupon sensitivity $\partial V_0 / \partial c \approx 0.30$ estimated from the same Monte Carlo engine, a coupon-equivalent gap of approximately 27 basis points (\$8,000 per \$10 million notional)—in the *protective* direction for the investor: a BSM-IV user underprices the snowball and, like the more-negative- ρ^{inv} case above, would tend to demand more compensation than the seller’s true model requires rather than be exploited.

9.4 Implications for Investor Protection

The information asymmetry identified here compounds the structural risk asymmetry documented in Sections 5 and 10. Even an investor who correctly identifies the three implicit option positions they are writing would remain at a pricing disadvantage if their model of the barrier probabilities is miscalibrated relative to the seller’s. The implied volatility surface is insufficient to close this gap. As Table 5 shows, the resulting rent is not large in absolute terms for a single notional—tens of basis points of coupon, a few thousand dollars per \$10 million—but it scales linearly with notional and, unlike the structural risk asymmetry of Section 5, is invisible to an investor who has correctly priced every vanilla option in sight.

Two regulatory implications follow. First, product providers should be required to disclose the model parameters used to set the coupon—including ρ , σ_v , and κ —since these govern the fair value of the embedded barrier options as directly as the contractual barrier levels themselves. Second, investors should understand that “fitting the market” in the sense of reproducing observed vanilla option prices does not guarantee accurate snowball pricing: any model with the wrong joint path dynamics will misassign probabilities across the KO, KI, and full-coupon scenarios, creating the scope for systematic mispricing documented here.

10 Risk Management Implications

10.1 The Snowball Effect in Dealer Hedging Books

The term “snowball effect” in structured-product markets refers to a self-reinforcing price dynamic that arises when dealers attempt to delta-hedge large outstanding snowball notionals. Because the snowball’s delta profile is positive and increasing as the spot falls toward the KI barrier, and positive but small and decreasing above the KO barrier, the dealer is effectively *short gamma* throughout the $[B_{KI}, B_{KO}]$ corridor. This means the dealer must buy the underlying as it falls and sell it as it rises—the opposite of a momentum strategy and the source of procyclical hedging pressure.

In a declining market, a large aggregate short-gamma position causes dealers to sell the underlying to maintain their delta hedge, which can amplify the initial price decline and potentially accelerate KI barrier breaches across many products simultaneously. This feedback loop is the titular snowball effect: a modest market decline grows into a self-reinforcing cascade as dealer hedging activity adds selling pressure, which triggers more KI events, which forces further de-risking.

10.2 Hedging Framework for the Note Seller

A financial intermediary that sells the snowball without taking directional risk must neutralise the embedded option exposures dynamically. The hedging programme follows directly from the payoff decomposition of Section 5 and connects to the model-risk results of Section 9.

Position decomposition. The seller’s liability decomposes into three components. First, a contingent coupon stream at rate c for as long as neither barrier has fired: since $c \gg r$, this is the dominant carry cost and the primary reason the seller welcomes early KO events. Second, a *short* series of binary digital calls at B_{KO} , one per observation date: each digital that fires terminates the liability with a fixed payment. Third, a *long* down-and-in put: in the KI-then-survive scenario the seller’s repayment obligation is

$$\min\left(\frac{S_T}{S_0}, 1\right) = 1 - \frac{(S_0 - S_T)^+}{S_0}, \quad (15)$$

so the seller pays *less* than par when $S_T < S_0$ after the knock-in—a built-in offset to the coupon liability that grows in value precisely when markets decline sharply.

Delta hedging. The seller maintains a long position in the underlying (typically index futures) to offset the net short-delta exposure of the note. The required hedge is largest and most rapidly changing as the spot approaches B_{KI} , consistent with the short-gamma characterisation above. At the moment a KI barrier is breached, the required delta falls sharply: the seller transitions from a large long-futures position (hedging the note’s elevated sensitivity above the barrier) to a much smaller one (the knocked-in down-and-in put has positive but lower delta for $S < S_0$ and near-zero delta for $S > S_0$ at maturity). This step-down is the proximate source of the concentrated selling pressure identified in the snowball effect: it is not the continuous delta rebalancing that creates the cascade, but the discontinuous unwinding triggered when S crosses B_{KI} across a large notional stock of outstanding

products.

Option overlay. Delta hedging alone leaves the seller exposed to large instantaneous moves and implied-volatility shifts. A standard two-layer overlay adds:

- *Put spreads near B_{KI} :* long an OTM put struck near $0.75 S_0$, short a further OTM put for cost reduction. These approximate the down-and-in put in (15); the residual path-dependence mismatch (vanilla vs. barrier-activated) is managed via intraday delta rebalancing.
- *Call spreads near B_{KO} :* long a call at $B_{KO} - \varepsilon$, short a call at $B_{KO} + \varepsilon$, to replicate each monthly digital. The half-width ε is calibrated to current implied volatility; tighter spreads capture more of the binary premium but increase local gamma at the autocall level.

Higher-order Greeks and model dependence. Vanna ($\partial^2 V / \partial S \partial \sigma$) is large near B_{KI} because spot and volatility co-move adversely in crash scenarios; the delta hedge must be adjusted simultaneously with the vega position, requiring a model that captures this joint dynamic. Volga ($\partial^2 V / \partial \sigma^2$) is material because the rescue effect depends nonlinearly on realised volatility—a feature absent from the BSM-IV benchmark. For worst-of or basket snowballs there is additionally a short-correlation exposure, typically hedged via dispersion trades or correlation swaps.

All these Greeks must be computed under a model that correctly captures the rescue mechanism. The information-asymmetry result of Section 9 demonstrates that two parameter vectors surface-equivalent at the ATM point can imply materially different KI and KO probabilities and therefore materially different hedge ratios. A seller using BSM-IV Greeks systematically underestimates gamma near B_{KI} , because BSM allocates mass more uniformly across paths while the Bates SVJ model concentrates KO exits in high-volatility episodes and leaves a thinner residual tail reaching the KI level. The hedging shortfall is largest in

the market conditions—elevated realised volatility, sharp downside—when accuracy matters most.

Gap risk and autocall clustering. Because the KI barrier is monitored continuously, an overnight gap through B_{KI} cannot be closed by delta hedging. Sellers typically pre-buy OTM puts just below B_{KI} as gap-risk insurance; the cost is ultimately embedded in the coupon offered to the investor. In a sustained bull market, autocall events cluster at the same monthly observation date across a large book of outstanding notes, forcing the seller to unwind multiple hedge positions simultaneously. The resulting demand in the options market—buying back put spreads, covering short call spreads—can temporarily compress implied volatility at the affected tenors and create short-lived dislocations that feed back into the delta-hedging process of the remaining live notes.

10.3 Implications for Product Design

Our findings suggest several practical considerations.

KO barrier placement. Issuers should recognize that raising the KO barrier marginally from 103% to 105% does not provide a meaningful benefit to investors (only 0.9 percentage points of additional full-coupon probability) but substantially increases KI risk. A KO barrier near 103–105% with monthly observation appears to price close to par under our calibration.

Observation frequency. The conventional monthly schedule appears to be a reasonable compromise. Annual monitoring generates far too much KI risk (30.6%) for a product investors typically believe to be conservative; the resulting above-par PV (1.053) suggests the coupon at annual frequency is too high relative to the embedded risk. Weekly monitoring eliminates much of the KI risk but delivers below-par PV, implying the coupon is insufficient compensation at that frequency.

Model risk. The relationship between model complexity and KI risk is subtler than a simple “richer model implies more tail risk” narrative. As Section 8 demonstrates, BSM parameterized by the market ATM implied volatility actually *overestimates* the marginal KI probability (16.6% vs. 14.9%) relative to Bates SVJ, because it misses the stochastic-vol clustering that rescues paths through early KO exits. However, this does not mean BSM is conservative: it simultaneously *underestimates* the pace of early autocalls and produces a smoother, less concentrated delta profile that understates the true gamma exposure near the KI barrier during high-vol regimes. Practitioners who use BSM-IV to construct delta hedges will therefore be systematically under-hedged for gamma precisely in the market conditions—elevated realized volatility, sharp downside moves—when the KI scenario is most imminent.

11 Conclusion

This paper has provided a unified pricing and risk analysis of snowball structured products under the Bates stochastic volatility with jumps model. Our principal contributions are as follows.

We showed that the snowball is structurally equivalent to a portfolio of a long risk-free bond, a short down-and-in put, and a short call at the KO level. The investor’s coupon is the aggregate premium for writing these two implicit options, and the investor’s implicit belief is that the probability of a 25%+ drawdown over the product’s life is small—a belief that our calibrated model suggests may be unduly optimistic.

We demonstrated that raising the KO barrier from 103% to 105% primarily transfers risk from the early-exit scenario to the KI loss scenario, with negligible gain in the probability of earning the full coupon at maturity. Investors who prefer a higher KO barrier believing it to be more protective are misled: the product simply keeps them exposed for longer.

We established a novel and counterintuitive result regarding observation frequency: more frequent KO checks lower the product’s fair value (by curtailing coupon accrual through

earlier autocalls) while simultaneously reducing the knock-in loss probability (by rescuing upward-trending paths before they can crash). This frequency-dependent risk-return trade-off should be central to the discussion between issuers and investors when structuring the observation schedule.

We benchmarked the Bates SVJ results against the Black–Scholes–Merton model parameterized by the 2-year ATM implied volatility (23.48%). Despite matching the ATM vanilla option price by construction, BSM-IV systematically underestimates the autocall frequency (82.0% vs. 83.8%) because it cannot replicate the stochastic-vol clustering that pushes paths through the low KO barrier; this leads BSM to overestimate residual KI exposure (16.6% vs. 14.9%) with direct implications for hedge accuracy. The right lesson is not that BSM uniformly over- or underestimates risk, but that it *misallocates* risk across scenarios—precisely in the conditions when tail events are most imminent.

We established that the implied volatility surface is an insufficient statistic for snowball pricing. The Bates parameter vector is not uniquely identified by vanilla option prices: in particular, the spot-variance correlation ρ , the volatility-of-volatility σ_v , and the mean-reversion speed κ can all vary substantially across surface-equivalent parameterisations while leaving the observable IV surface unchanged. A seller who knows the true ρ —which governs the rescue mechanism documented in Section 8—and a buyer who calibrates to a surface-equivalent ρ^{inv} that is more negative than the truth will assign different probabilities to the KI and KO scenarios. The buyer with an overstated leverage effect will overestimate the rescue probability, underestimate KI risk, and overprice the snowball—accepting a coupon below fair value believing it to be adequate. The informational rent accruing to the seller is invisible from the public implied volatility surface.

Finally, the delta analysis confirms that snowballs impose a structurally short-gamma position on dealers across the $[B_{\text{KI}}, B_{\text{KO}}]$ corridor, generating the procyclical hedging dynamics underlying the snowball effect in equity markets.

Future research could extend this analysis in several directions: studying how the snow-

ball's fair value and scenario probabilities respond to volatility regime changes; analyzing multi-underlying snowball products (baskets or worst-of structures); empirically calibrating the model to snowball-implied prices from secondary markets; and quantifying the P&L impact of BSM-IV hedging errors relative to a Bates SVJ benchmark hedge.

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Figures

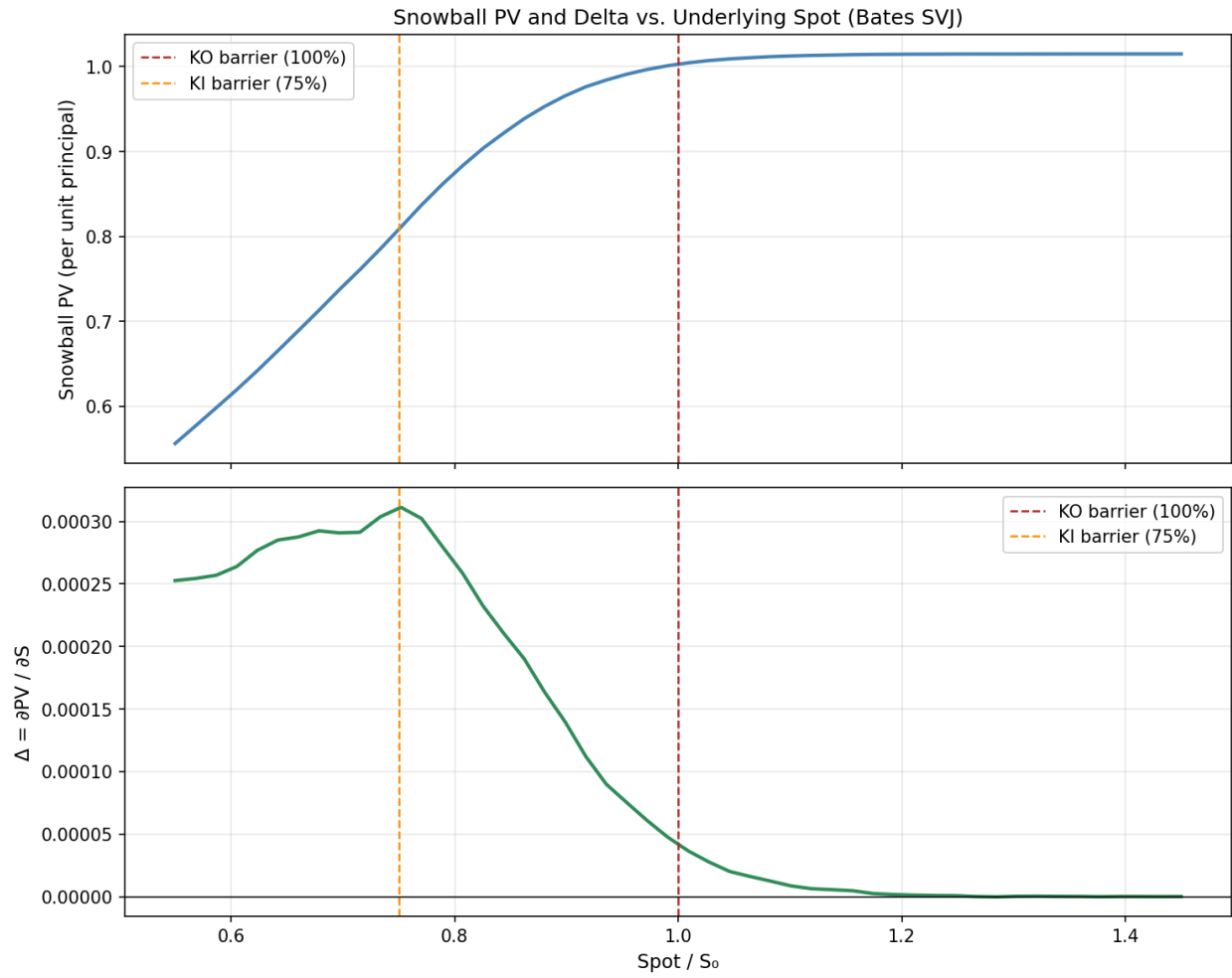


Figure 1: Snowball PV (upper panel) and delta (lower panel) as functions of the current spot normalized by the initial price S_0 . Parameters: KO = 100%, KI = 75%, coupon = 20%/yr, $T = 2$ years, monthly KO observation, Bates SVJ model. The vertical dashed lines mark the KI (orange) and KO (red) barriers. Delta peaks just above the KI barrier and collapses to near zero above the KO barrier.

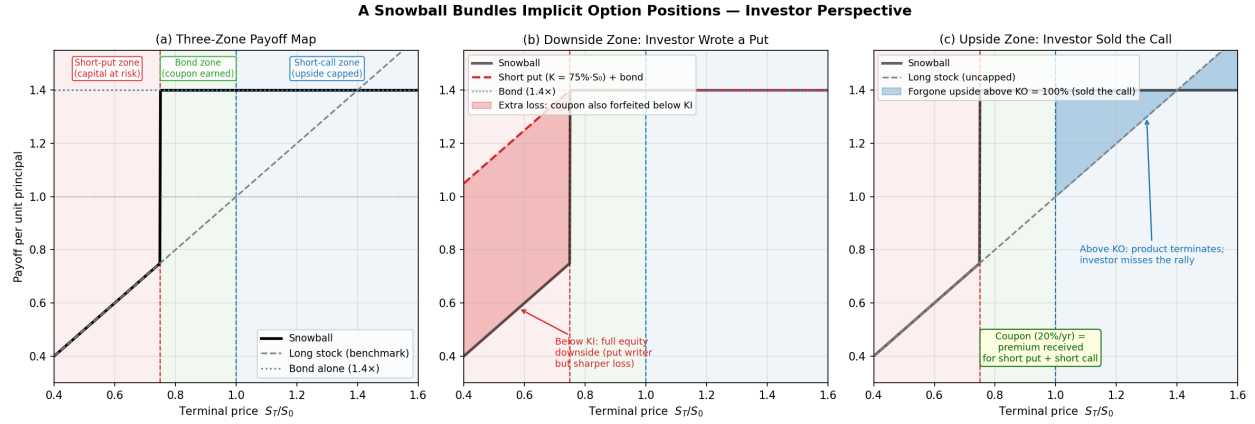


Figure 2: Snowball payoff decomposition. Panel (a) shows the simplified at-maturity payoff in three zones; Panel (b) compares the snowball to a short put plus bond, isolating the downside risk component; Panel (c) compares the snowball to a long-stock benchmark, shading the forgone upside above the KO barrier. The coupon equals the combined premium for the short put and the short call.

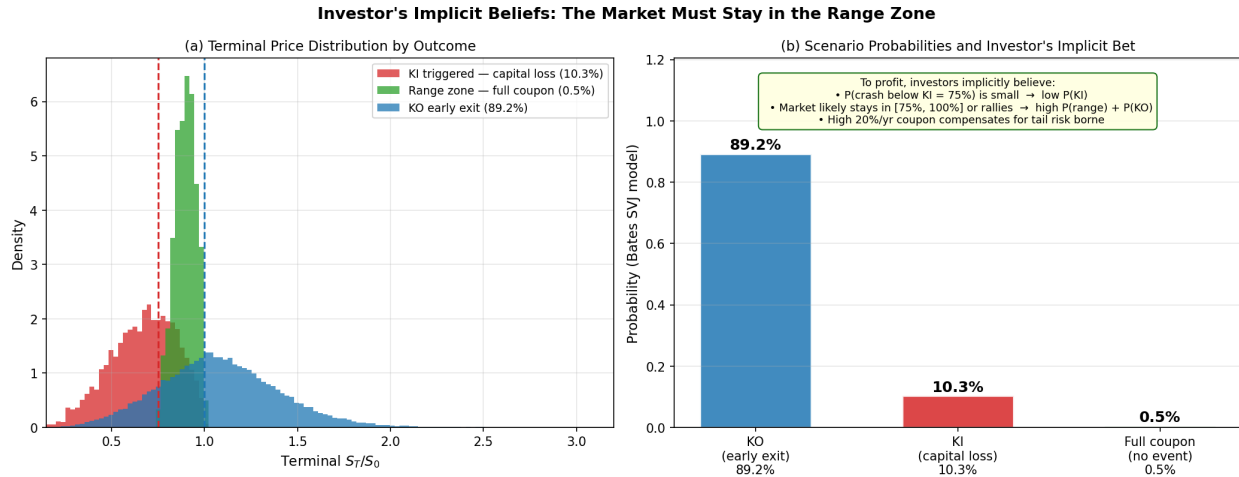


Figure 3: Investor's implicit beliefs under the Bates SVJ model. Panel (a) shows the distribution of terminal prices S_T/S_0 , colored by outcome (KI loss in red, range-zone coupon in green, KO early exit in blue). Panel (b) reports the unconditional probabilities of each scenario and summarizes the three conditions the investor must believe to expect a positive expected profit.

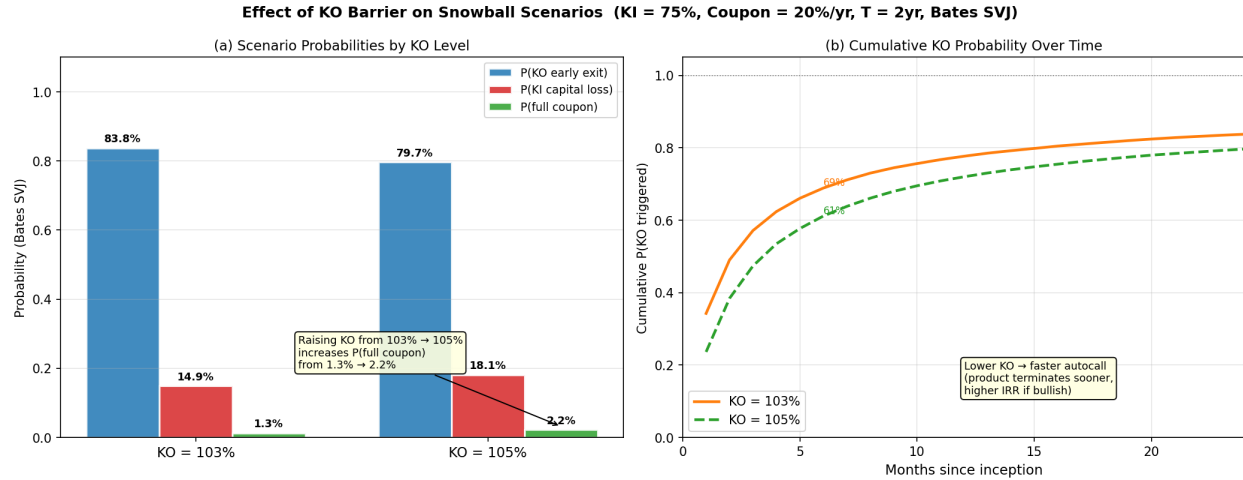


Figure 4: Effect of the KO barrier level on scenario probabilities. Panel (a): grouped bar chart comparing $P(KO)$, $P(KI)$, and $P(\text{full coupon})$ for $KO = 103\%$ and $KO = 105\%$. Panel (b): cumulative KO probability over the 24 monthly observation dates for each KO level. Common random numbers ensure the comparison is exact.

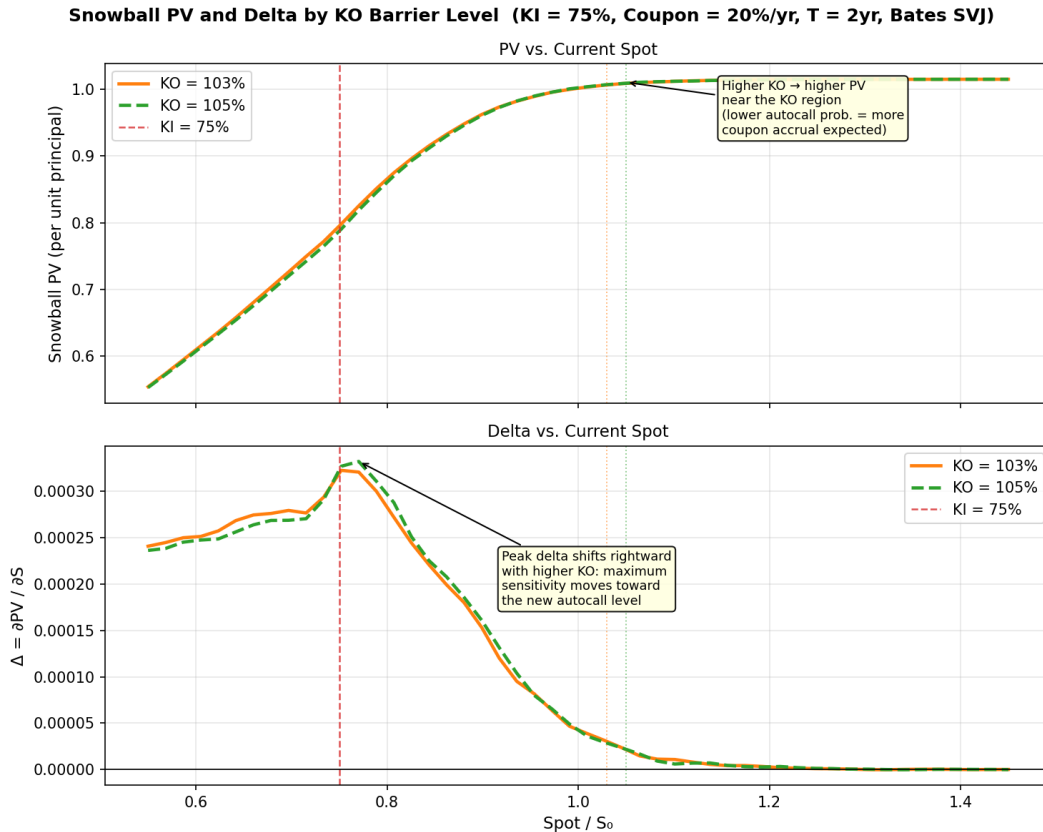


Figure 5: Effect of the KO barrier level on PV and delta curves. Upper panel: snowball PV as a function of current spot for $KO = 103\%$ (orange) and $KO = 105\%$ (green). Lower panel: corresponding delta curves. Both products price near par at $S = S_0$. Dotted vertical lines mark the respective KO barriers.

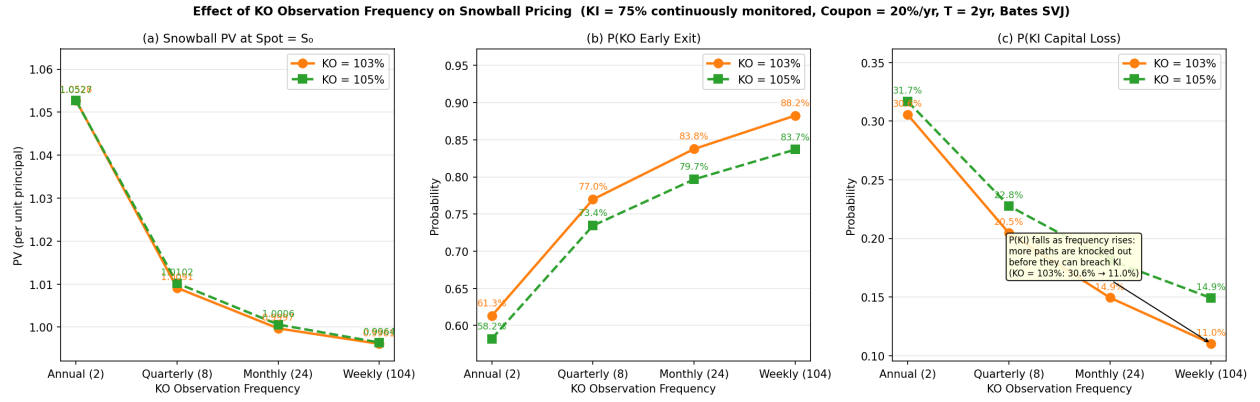


Figure 6: Effect of KO observation frequency on snowball pricing. Panel (a): fair value at $S = S_0$ falls as the observation schedule becomes more frequent. Panel (b): $P(\text{KO early exit})$ rises monotonically with frequency. Panel (c): $P(\text{KI capital loss})$ falls as frequent observation rescues upward-trending paths before they can crash through the barrier. All curves use the same simulated paths (common random numbers).

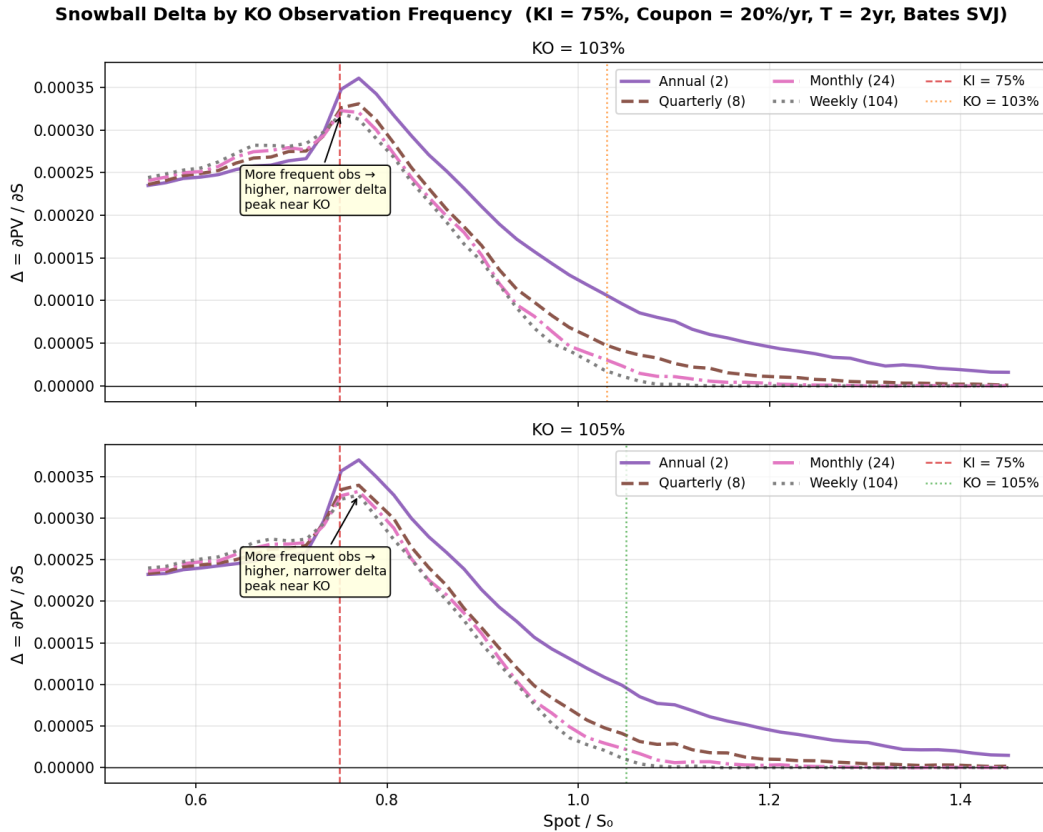


Figure 7: Snowball delta curves by KO observation frequency for $\text{KO} = 103\%$ (upper panel) and $\text{KO} = 105\%$ (lower panel). More frequent observation produces a taller, narrower delta peak near the KO barrier and faster decay above it.

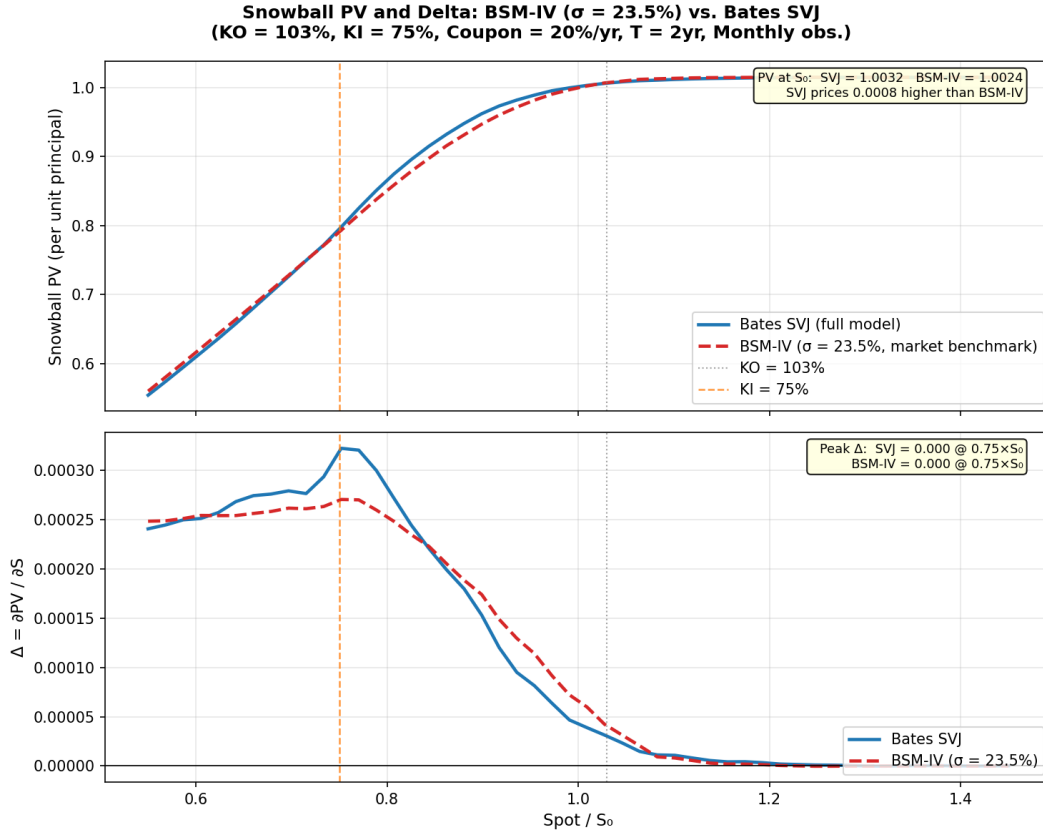


Figure 8: BSM-IV vs. Bates SVJ: PV (upper panel) and delta (lower panel) as functions of current spot. The BSM-IV model uses $\sigma_{IV} = 23.48\%$, the 2-year ATM implied volatility backed out from the Bates SVJ ATM call price. Both models agree at $S = S_0$ by construction. The Bates SVJ delta peak is taller and more concentrated near the KI barrier, reflecting the gamma concentration generated by stochastic-vol clustering. Parameters: KO = 103%, KI = 75%, coupon = 20%/yr, $T = 2$ yr, monthly obs.

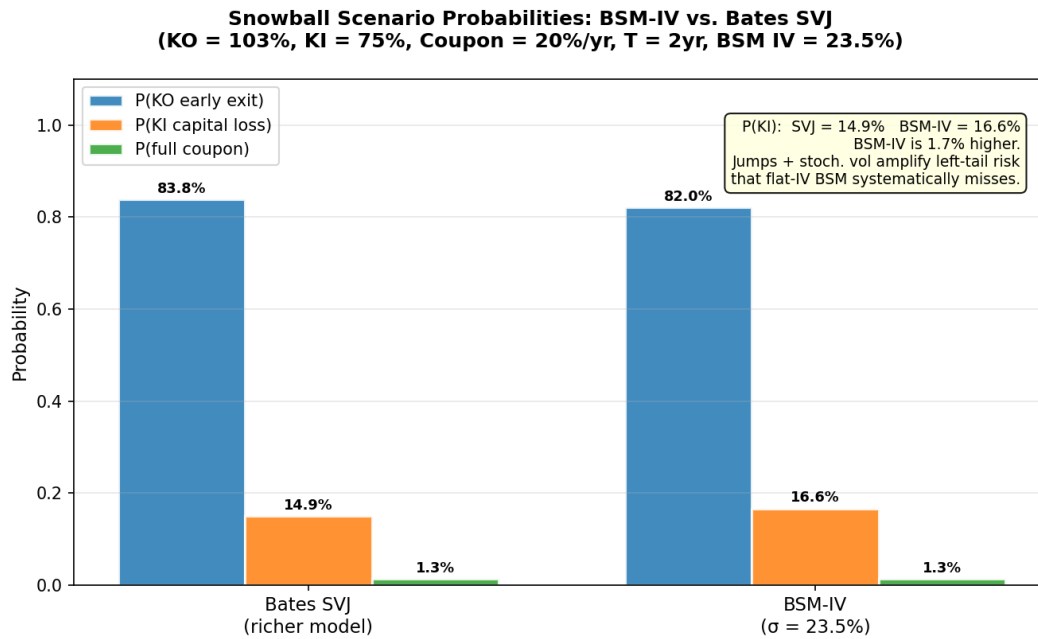


Figure 9: Scenario probability comparison: Bates SVJ vs. BSM-IV. Grouped bars show $P(\text{KO early exit})$, $P(\text{KI capital loss})$, and $P(\text{full coupon})$ for each model. Bates SVJ generates a higher autocall probability (83.8% vs. 82.0%) due to stochastic-vol clustering, while BSM-IV overestimates the residual KI exposure (16.6% vs. 14.9%). Parameters as in Table 4.